Figure 9.2: Typical Wave Packet for a Nonrelativistic Free Particle. Top: The initial wave packet is compact and highly localized. Bottom: Over time, the wave packet moves to the right and spreads out.

9.3 TIME-INDEPENDENT SCHRÖDINGER EQUATION

We are going to solve the time-dependent Schrödinger equation for nonrelativistic free particles, but first we need to solve the time-independent version. The time-independent equation is essentially the eigenvector equation for the Hamiltonian,

$$H|\Psi\rangle = E|\Psi\rangle,$$

written explicitly in terms of the wave function $\psi(x)$:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} = E\psi(x). \tag{9.5}$$

It's very easy to find a complete set of eigenvectors that satisfy this equation. In fact, momentum eigenvectors do the job. Let's try the function

$$\psi(x) = e^{\frac{ipx}{\hbar}} \tag{9.6}$$

as a possible solution. Carrying out the derivatives, we find that this function is indeed a solution to Eq. 9.5, as long as we set

$$E = p^2/2m. \tag{9.7}$$

This should come as no surprise—after all, E represents an energy eigenvalue in Eq. 9.5.

Exercise 9.1: Derive Eq. 9.7 by plugging Eq. 9.6 into Eq. 9.5.